

Lab 3 · Week 1 · Teacher Guide

Fix the Lock · Sundays 3:30 to 5:30 · Aasees Sethi · Caution Tape

What they learn today

- Your game runs on your laptop, so players can change anything in it, including checks.
- Scores live on Google's computers. That's where the "is this allowed?" decision belongs.
- Security Rules are a bouncer: not on the list, not getting in. It checks the uid, never the name.
- Rules must cover both scores and the runs folder inside it.

Your coach build: Skyline

- You hack your own city live at the start, then lock it live before they do.
- Line to repeat all term: your laptop asks, Google decides.

Before class

- Skyline running locally WITH the console door, City open in a second window, hack tested.
- skyline-starter repo is public with SETUP.md. Write the link on the board.
- Test the rules prompt on your Skyline at home so the coach build publishes cleanly.
- Check which kids have a Google account on a parent's email (needed for Firebase).

Slide controls

- F: full screen · Right arrow or clicker: next · Left arrow: back.
- S: start or pause the timer · R: reset it.
- Interactive slides: click Send, Delete the check, Bouncer ON/OFF. Quizzes: click the answer a kid picks.

Run of show

#	Time	What happens
1	3:30 to 3:40	Welcome back
2	3:40 to 3:42	Ever cheated in a game?
3	3:42 to 3:44	Meet Skyline
4	3:44 to 3:45	Our class city
5	3:45 to 3:46	Today's missions
6	3:46 to 3:50	Watch this
7	3:50 to 4:00	Build your city, part 1
8	4:00 to 4:08	Build your city, part 2
9	4:08 to 4:15	Build your city, part 3
10	4:15 to 4:23	Mission 2: hack your city
11	4:23 to 4:26	Why did that work?
12	4:26 to 4:30	"Just add a check in the game!"
13	4:30 to 4:33	Now try to hack this one

#	Time	What happens
14	4:33 to 4:36	Quiz 1 of 3
15	4:36 to 4:46	Snack break
16	4:46 to 4:49	Mission 3: Security Rules = a bouncer
17	4:49 to 4:53	Put the bouncer to work
18	4:53 to 4:56	What's on the bouncer's list
19	4:56 to 4:59	Quiz 2 of 3
20	4:59 to 5:01	Heads up: 2 things get saved
21	5:01 to 5:05	Coach build: lock my city
22	5:05 to 5:15	Your turn: hire your bouncer
23	5:15 to 5:20	Hack it one more time
24	5:20 to 5:22	Close the back door
25	5:22 to 5:26	Quiz 3 of 3
26	5:26 to 5:30	Say it together + next week

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Welcome back, builders

Today you're going to hack a game.
Then you're going to stop yourself. 😊



Slide 1 of 26

3:30 to 3:40 • 10 min

Welcome back

Goal: Kids settle. Returning kids reconnect, new kids get a buddy.

Say: Welcome back, builders! Today you're going to hack a game. Then you're going to stop yourself.

Do: Say hi to returning kids by name. Pair every new kid with someone from last term. Press the right arrow to reveal the second line.

Kids do: Open laptops, VS Code, and sign in to Codex.

If stuck: A kid's parent isn't there to approve anything: note it, you'll need them for the Firebase Google account.

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Ever cheated in a game?

Glitches count. Cheat codes count. No judging.

Slide 2 of 26

3:40 to 3:42 • 2 min

Ever cheated in a game?

Goal: Hook. Get them talking about cheating so the lesson feels like theirs.

Say: Ever cheated in a game? Glitches count. Cheat codes count. No judging.

Do: Let 2 or 3 kids share. Laugh with them.

Kids do: Share stories.

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Meet Skyline

A crane swings a floor. You drop it. Stack as high as you can.

Your best score becomes a building in our class city.

► Open Skyline



Slide 3 of 26

3:42 to 3:44 • 2 min

Meet Skyline

Goal: They meet the game they'll be hacking today.

Say: This is Skyline. I made it. A crane swings a floor, you drop it, you stack as high as you can. Your best score becomes a building in our class city.

Do: Click Open Skyline and play one quick round on the projector.

Kids do: Watch.

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Our class city



Slide 4 of 26

3:44 to 3:45 • 1 min

Our class city

Goal: They see what a leaderboard looks like when it's a city.

Say: Every building is a player. The crown is the top score. The YOU light is you.

Do: Point at the crown and the YOU beacon. Point at the two aasees buildings and say: hold that thought, it matters later.

Kids do: Look.

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Today's missions



Mission 1
Build your city



Mission 2
Hack it



Mission 3
Hire a bouncer

By the end, the same hack bounces right off your game. 

Slide 5 of 26

3:45 to 3:46 · 1 min

Today's missions

Goal: They know the 3 parts of today.

Say: Mission 1, build your own city. Mission 2, hack it. Mission 3, hire a bouncer so nobody can hack it again.

Do: Point at each card. Every later slide has a mission tag.

Kids do: Listen.

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Watch this.

I'm not going to play. I'm just going to type one line:

```
app.submitScore(999999)
```

...and the game just believed me. 

Slide 6 of 26

3:46 to 3:50 · 4 min

Watch this

Goal: The hook: you cheat live and they see it work.

Say: I'm not going to play. I'm just going to type one line. (Type it.) And the game just believed me.

Do: Before revealing anything: open Skyline and the City, press F12, Console, type `app.submitScore(999999)`, Enter. Show your building blast off. Then come back and press the right arrow.

Kids do: React. Ask how you did it.

If stuck: If you get `app` is not defined, your deployed Skyline has no console door. Run it locally with the door added.

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Mission 1

Build your city, part 1

Step 1
Get the code
In the VS Code terminal:
`git clone <link on board>`
`cd skyline-starter` then `npm install`

Step 2
Make a Firebase project
`console.firebase.google.com`
Add project, call it skyline, turn off Analytics.

Already have a working leaderboard from last term? You can use yours! 🚀

10:00 Tap to start

Slide 7 of 26

3:50 to 4:00 · 10 min

Build your city, part 1

Goal: Every kid has the Skyline code and their own Firebase project.

Say: Mission 1. Everyone gets their own city. First, the code. Then your own Firebase project.

Do: Write the skyline-starter repo link on the board. Do each step on the projector. Press S for 10 minutes. Returning kids with a working leaderboard can use theirs.

Kids do: git clone, cd skyline-starter, npm install. Create a Firebase project called skyline, Analytics off.

If stuck: Firebase needs a Google account made on a parent's email. No account: pair them up and follow up with the parent after class.

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Mission 1

Build your city, part 2

Step 3
Turn on sign-in
Authentication, Sign-in method.
Switch ON Google and Anonymous. Two switches!

Step 4
Make the database
Firestore Database, Create. Rules tab: paste the open rules from SETUP.md and Publish.

These rules say "anyone can do anything". Yep. That's the problem we fix today.

8:00 Tap to start

Slide 8 of 26

4:00 to 4:08 · 8 min

Build your city, part 2

Goal: Sign-in turned on, database created with the open rules.

Say: Two switches: Google AND Anonymous. Everyone forgets the second one. Then make the database and paste the open rules. These rules say anyone can do anything. That's the problem we fix today.

Do: Show each click on the projector. Press S for 8 minutes.

Kids do: Enable Google and Anonymous. Create Firestore in production mode. Paste the open rules from SETUP.md and Publish.

If stuck: Guest sign-in fails later: Anonymous is off.

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Mission 1

Build your city, part 3

1. Firebase: Project settings, Your apps, `</>`, copy the config
2. Paste it into `src/firebase-config.js`
3. Terminal: `npm run dev`
4. Sign in as a guest and play one round

👍 when your building shows up in the city!

Skyline

Stack a tower. Build the city.

Continue with Google

or

Your name

Play as guest

7:00 Tap to start

Slide 9 of 26

4:08 to 4:15 • 7 min

Build your city, part 3

Goal: Every kid's Skyline runs and their building appears.

Say: Copy your config, paste it into `firebase-config.js`, run `npm run dev`, sign in as a guest and play one round. Thumbs up when your building shows up.

Do: Press S for 7 minutes. Check thumbs.

Kids do: Paste config, `npm run dev`, play.

If stuck: Almost there screen: the config isn't pasted right, check the quotes. `auth/unauthorized-domain`: they're not on localhost.

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Mission 2

Your turn: hack your city

1. Press F12, click Console (Mac: `Cmd + Option + J`)
2. Type `app.submitScore(999999)` and hit Enter
3. Look at the city. Who's #1 now? 😊
4. Screenshot it. That's your BEFORE picture 📸

Got "app is not defined"? Ask Codex: Add a temporary line so I can call my `submitScore` function from the browser console as `app.submitScore(number)`.

8:00 Tap to start

Slide 10 of 26

4:15 to 4:23 • 8 min

Mission 2: hack your city

Goal: Every kid cheats their way to #1 and screenshots it.

Say: Mission 2! Your turn. F12, Console, type the line, hit Enter. Then look at your city. Screenshot it, that's your BEFORE picture.

Do: Press S for 8 minutes. Walk around. Make sure every kid gets to #1.

Kids do: Hack, screenshot.

If stuck: `app is not defined`: press the right arrow to show the Codex prompt that adds the console door.

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Mission 2

Why did that work? 🤔



Your laptop

The game runs here. You can change ANYTHING here.

⇌



Google's computers

Far away. They save the scores. You can't touch them.

 The game decided your score on YOUR laptop. So you could lie to it.

Slide 11 of 26

4:23 to 4:26 · 3 min

Why did that work?

Goal: The big idea of the whole term: your laptop vs Google's computers.

Say: Your game runs on YOUR laptop. You can change anything there. The scores live on Google's computers, far away. The game decided your score on your laptop, so you could lie to it.

Do: Press the right arrow for each box, then the line.

Kids do: Listen, answer questions.

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Mission 2

"Just add a check in the game!" Let's try 🧪

Your laptop

```
function submitScore(score) {  
  if (score > 3000) return // no cheating!  
  save(score)  
}
```

 Delete the check

=====

🏆 Leaderboard

Maya	1,420
Sam	980
Leo	610

Slide 12 of 26

4:26 to 4:30 · 4 min

"Just add a check in the game!"

Goal: They see why a check inside the game never works.

Say: Someone always says: just add a check in the game! Let's try it.

Do: Click Send with 999999: the game blocks it. Then click Delete the check and Send again: it lands at #1. Say: that's what a hacker does. It's your laptop, you can delete anything. Let a kid drive.

Kids do: Guess what happens each time.

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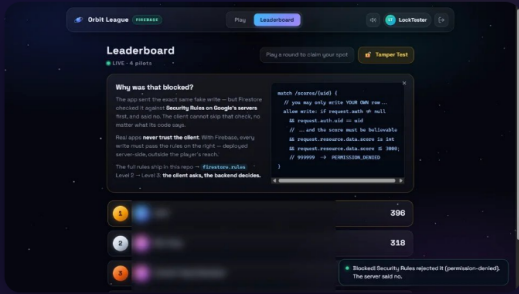
Mission 2

Now try to hack this one

demo-leaderboard.ct839.dev

1. Type a name, hit Launch
2. Click Leaderboard
3. Click Tamper Test

It has a **bouncer**. 🕶️



Slide 13 of 26

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Quiz 1 of 3 Class score: 1/3 ★

1. Where does your game actually run?

☒ A On my laptop
 ☐ B On Google's computers
 ☐ C On the teacher's laptop

The game runs on your laptop. That's why you can mess with it. 🖥️
2. Why can't a check inside the game stop cheaters?

☐ A It's too slow
 ☐ B Players can delete it
 ☐ C It only works on Macs
3. So where should the "is this score allowed?" check live?

☐ A In the game
 ☐ B In the browser
 ☐ C On Google's computers

Slide 14 of 26

4:30 to 4:33 • 3 min

Now try to hack this one

Goal: They feel the difference: a game with a bouncer.

Say: Same hack, different game. Try it.

Do: Kids open demo-leaderboard.ct839.dev, Launch, Leaderboard, Tamper Test. Right arrow shows the blocked screenshot, again shows: it has a bouncer.

Kids do: Try Tamper Test, get blocked.

4:33 to 4:36 • 3 min

Quiz 1 of 3

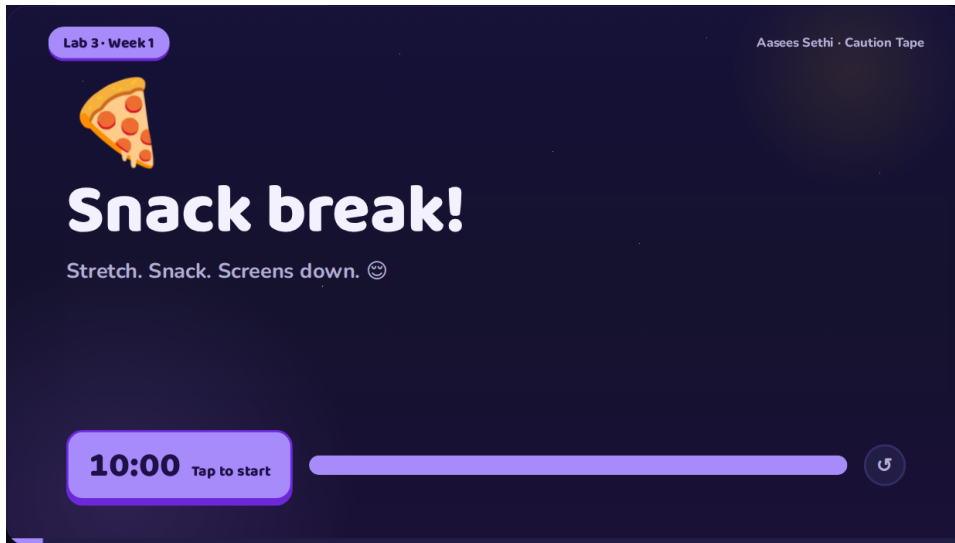
Goal: Check they got laptop vs Google's computers.

Say: Quiz time!

Do: Read each question, kids shout A, B or C, one kid clicks.

Kids do: Answer.

If stuck: Answers: 1 A (my laptop). 2 B (players can delete it). 3 C (on Google's computers).



Slide 15 of 26

4:36 to 4:46 • 10 min

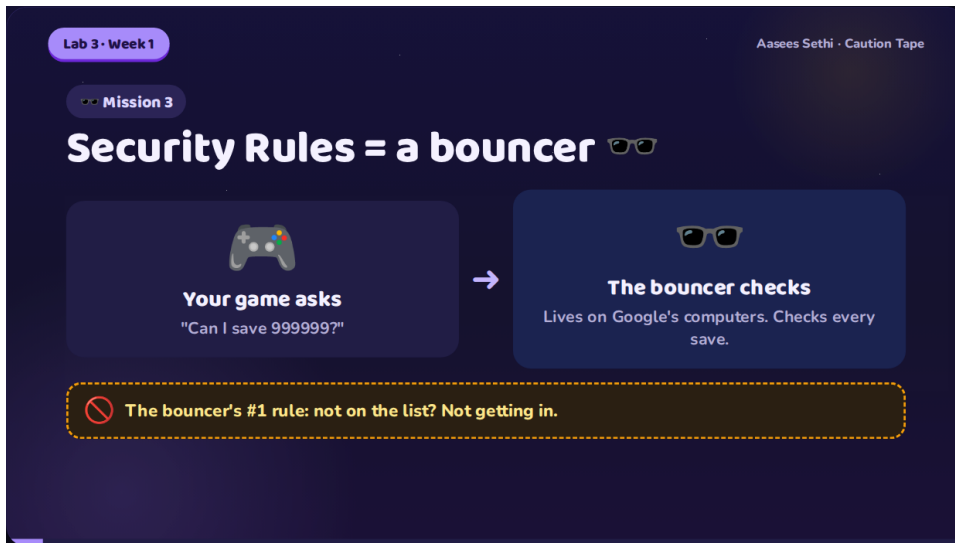
Snack break

Goal: Rest.

Say: Snack break! Screens down.

Do: Press S for 10 minutes. Help anyone stuck on setup.

Kids do: Snack.



Slide 16 of 26

4:46 to 4:49 • 3 min

Mission 3: Security Rules = a bouncer

Goal: They understand Security Rules as a bouncer on Google's computers.

Say: Mission 3. Security Rules are like a bouncer. Your game asks: can I save 999999? The bouncer checks. It lives on Google's computers, so you can't hack it from the console. And its number one rule: not on the list, not getting in.

Do: Press the right arrow for the last line. Say it twice.

Kids do: Listen.

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Mission 3

Put the bouncer to work

 **Your laptop**

```
function submitScore(score) {  
  save(score) // no check at all  
}
```

Zero checks here. Try to cheat!

 **Google's computers**

Bouncer:

Maya	1,420
Sam	980
Leo	610

Slide 17 of 26

4:49 to 4:53 • 4 min

Put the bouncer to work

Goal: They see the bouncer block cheats but let honest scores in.

Say: Zero checks in the game this time. Watch.

Do: Bouncer OFF: send 999999, it gets in. Click to turn it ON: send 999999, it bounces back. Type 1500, it saves. Type 12.5, it bounces (not a whole number).

Kids do: Guess each result before you click.

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Mission 3

What's on the bouncer's list

allow read: if true	 Anyone can look at the city
request.auth.uid == uid	 You can only save YOUR building
score is int	 Whole numbers only
score ≤ 3000	 No crazy scores
allow delete: if false	 Nobody deletes anything

Slide 18 of 26

4:53 to 4:56 • 3 min

What's on the bouncer's list

Goal: They can read each rule in plain English.

Say: Here's the list. Anyone can look at the city. You can only save YOUR building. Whole numbers only. No crazy scores. Nobody deletes.

Do: Read line by line. On the uid line, point back at the two aasees buildings: same name, two different people. The bouncer checks the uid, never the name.

Kids do: Say each line back in their own words.

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? Quiz 2 of 3 Class score: 1/3 ★

Would the bouncer let it in? 🕶️

1. You're signed in, saving YOUR score of 1,200
A ✓ Let in B ✗ Kick out
Signed in, your own building, a real score. Welcome in! ✓
2. Changing SOMEONE ELSE's score
A ✓ Let in B ✗ Kick out
3. Saving a score of 12.5
A ✓ Let in B ✗ Kick out

Slide 19 of 26

4:56 to 4:59 • 3 min

Quiz 2 of 3

Goal: Practice thinking like the bouncer.

Say: Would the bouncer let it in?

Do: Kids shout Let in or Kick out, one kid clicks.

Kids do: Answer.

If stuck: Answers: 1 Let in. 2 Kick out (not your uid). 3 Kick out (not a whole number).

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🔍 Mission 3

Heads up! Your game saves 2 things 📁

📁 scores

📄 your building your best score ✓ on the list

📁 runs

📄 every game you play run 1, run 2, run 3... ✗ forgot = kicked out

⚠️ The bouncer needs a rule for BOTH. Don't forget the runs!

Slide 20 of 26

4:59 to 5:01 • 2 min

Heads up: 2 things get saved

Goal: Avoid the most common mistake of the lesson.

Say: Your game saves two things: your best score, and every game you play in a folder called runs. The bouncer needs a rule for BOTH.

Do: Right arrow: green tag, red tag, then the warning.

Kids do: Listen.

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🔗 Watch me lock my city

I'm hiring a bouncer for Skyline 🕶️

I ask Codex for the rules, then read every line before I trust it:

```
Create firestore.rules. Anyone can read scores. A signed-in player may create or update only scores/{their uid} and add runs inside it. Score must be a whole number from 0 to 3000. Nobody deletes. Explain each line.
```

1 Paste in Firebase, Publish

2 Hack it again

3 Blocked! 🚫

Slide 21 of 26

5:01 to 5:05 · 4 min

Coach build: lock my city

Goal: They watch you do the whole fix once before they do it.

Say: Watch me hire a bouncer for Skyline. I ask Codex for the rules, and I read every line before I trust it.

Do: In your Skyline project, send the prompt to Codex. Read the explanation out loud. Paste into Firebase Rules, Publish. Hack it again live: red error. Celebrate.

Kids do: Watch.

If stuck: Test this at home first so you know it publishes cleanly.

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🎯 Mission 3

Your turn: hire your bouncer 📝

1. Send the same prompt to Codex. Change 3000 to your real best score
2. Read what each line does. Can you explain it? 😊
3. Firebase, Firestore, Rules tab. Delete the old rules, paste yours, **Publish**
4. Data tab: delete your 999999 building ✅

Red error when you publish? Usually a missing ; or }. Paste it into Codex.

10:00 Tap to start

↺

Slide 22 of 26

5:05 to 5:15 · 10 min

Your turn: hire your bouncer

Goal: Every kid writes, reads and publishes their rules.

Say: Your turn. Same prompt. Change 3000 to your real best score. Read every line, can you explain it? Then publish, and delete your 999999 building.

Do: Press S for 10 minutes. Ask a few kids to explain one line to you.

Kids do: Codex prompt, read, publish, delete 999999.

If stuck: Red error on publish: usually a missing ; or }, paste it into Codex. "Why could I delete it?" Because the console is the owner's tool. The bouncer doesn't stop the owner.

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Mission 3

Hack it one more time

```
app.submitScore(999999)
```

✗ Missing or insufficient permissions

Red error = good news! Your bouncer said no. 🎉

Now play one real round. Your building should still grow. 🏠

5:00 Tap to start ↻

Slide 23 of 26

5:15 to 5:20 • 5 min

Hack it one more time

Goal: The payoff: the hack fails, honest play still works.

Say: Try the hack again. See that red error? That's good news. Your bouncer said no. Now play one real round, your building should still grow.

Do: Right arrow three times as kids hit each moment. Press S for 5 minutes.

Kids do: Hack (blocked), then play honestly (saved).

If stuck: Honest score blocked too: the rules are stricter than what the game sends. Paste the Console error and rules into Codex and ask which rule blocked it. Usually the runs rule is missing.

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Mission 3

Close the back door

```
Remove the temporary app.submitScore console line.
```

1. Screenshot the red error. That's your AFTER picture 📸
2. Put BEFORE + AFTER in a folder called security
3. Commit and push ✅

Slide 24 of 26

5:20 to 5:22 • 2 min

Close the back door

Goal: Remove the console door and save their work.

Say: Last step. Remove the back door, screenshot your red error, and push your work.

Do: Make sure every kid sends the remove prompt. Don't skip it.

Kids do: Remove the console line, save BEFORE and AFTER screenshots in a security folder, commit and push.

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? Quiz 3 of 3 Class score: 1/3 ★

1. Where does the bouncer live?

☐ A In my game ☒ B On Google's computers ☐ C In VS Code

The bouncer (Security Rules) lives on Google's computers. ☁

2. What do people forget to make a rule for?

☐ A The runs ☐ B The city ☐ C The crane

3. "Missing or insufficient permissions" today means...

☐ A My game is broken ☐ B Firebase is down ☐ C My bouncer worked

Slide 25 of 26

5:22 to 5:26 • 4 min

Quiz 3 of 3

Goal: Lock in the 3 big ideas.

Say: Last quiz!

Do: Same as before. Run the official live quiz here instead if you have it.

Kids do: Answer.

If stuck: Answers: 1 B (Google's computers). 2 A (the runs). 3 C (my bouncer worked).

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Say it with me:

**Your laptop asks.
Google decides.** 🕶

Next week: Poké Arena, where the server decides everything! 🎮

👜 **Bring:**

- GitHub and Vercel logins
- Your favourite Gen 1 Pokémon

Slide 26 of 26

5:26 to 5:30 • 4 min

Say it together + next week

Goal: End with the line of the term.

Say: Say it with me: your laptop asks, Google decides. Next week, Poké Arena, where the server decides everything. Bring your GitHub and Vercel logins and your favourite Gen 1 Pokémon.

Do: Say the line together twice. Check everyone pushed.

Kids do: Say it, pack up.